Code and Branch Coverage Analysis Report

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Project	Daycare Fees
Application File	daycare_fees.py
Test File	test_daycare_fees.py
Testing Tools	Python 3.12.4, pytest 9.2.0, coverage 7.x

1. Objective

The objective of this lab is to measure the completeness of the tests for the daycare fees application, identify untested statements and decision branches, and add tests until both code coverage and branch coverage exceed the required 90% threshold.

2. Application Overview

The application contains three functions. **extra_hours_fee()** validates the age group and hours, calculates included hours and any extra-hours fee, and applies the appropriate rate. **enrolment_allowed()** determines eligibility using immunisation status, age, and vacancies. **pickup_status()** determines whether pickup is on time, late, or overdue based on the remaining minutes, aftercare, and Friday cutoff rules.

3. Test Design

• Extra-hours fee: included-hour boundaries, sibling allowance for eligible age groups, positive extra hours, higher-rate thresholds, unknown age groups, and negative hours.

• Enrolment: pending immunisation with ages above and below 12 months; minimum-age boundary; no vacancies; and available vacancies.

• Pickup status: on-time, late, and overdue outcomes for standard pickup and aftercare, including Friday cutoff adjustments.

4. Coverage Results

Metric	Result	Details
Test execution	28 passed	All collected tests passed
Statement/code coverage	100%	37 statements covered; 0 missed
Branch coverage	100%	26 branches; 0 partial branches

The recorded terminal output showed 28 tests passing. The regular coverage report showed 37 statements, 0 missed, and 100% coverage. The branch-enabled report showed 26 branches, 0 partial branches, and 100% coverage.

5. Commands Used

```
python3 -m pytest test_daycare_fees.py -v
```

```
python3 -m coverage erase
python3 -m coverage run -m pytest test_daycare_fees.py
python3 -m coverage report -m daycare_fees.py
python3 -m coverage erase
python3 -m coverage run --branch -m pytest test_daycare_fees.py
python3 -m coverage report -m daycare_fees.py
```

6. Conclusion

The expanded test suite exercises the application’s statement logic and decision branches. Based on the captured terminal results, both code coverage and branch coverage reached 100%, exceeding the lab requirement of at least 90%. The two terminal screenshots should be included with the source and test files in the GitHub repository.

7. Submission Checklist

■ Include this analysis report in the repository.

■ Include test_daycare_fees.py and daycare_fees.py.

■ Include code_coverage.png and branch_coverage.png.

■ Confirm that the two coverage commits and all deliverables are pushed to the repository required by the instructor.